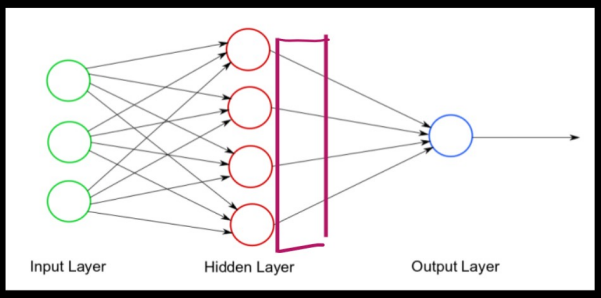
Transformer



NN

word2vec

intermediate o/p of NL

machine translation

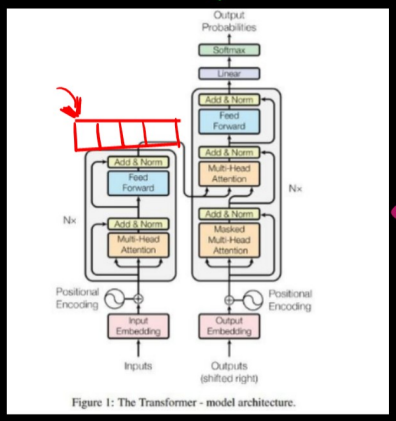


Figure 1: The Transformer - model architecture.

① text classification
② Summarization
③ translation

④ text generation

NLP

SOTA
state of Art
LLM (GPT, Gemini, HF)
AI

Embedding
intermediate O/P

Prompt → gpt → text classification (Pos|Neg, Spam|Ham)
→ Summarization
→ translation (Eng → hindi)

(transformer decoder)

my name is Sunny

① Pretraining
(Next token)

Previous sent → Next token

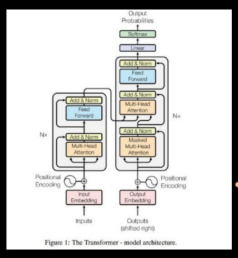
① 2017-19 → transformer → Machine translation
→ text classification =
→ Summarization =

② BERT → Google → MLM

Sunny is [MASK] mentor
masked

③ gpt → OpenAI → Autoregressive training

Sunny is AI mentor.
next token



LLM (text generation)

(embedding model)

convert our data into meaning numbers

(Numbers) embedding

1. word
2. sent-
3. Para/document-
4. Img
5. Audio
6. video

(transformer) (embedding)

NN → word2vec (HL)

SelfAttention → FF

SOTA → transformer → SOTA embedding model → meaningful numbers

Any type of data

Feature	Encoding (One-hot / TF-IDF)	Neural Network Embedding (Word2Vec)	Transformer Embedding (OpenAI, Gemini, Sentence Transformer Based Model)	Reason
Meaning capture	No semantics	Some semantics	Deep contextual semantics	Encoding is frequency-based, NN learns patterns, Transformer understands context
Context awareness	None	Limited (window-based)	Full context (bidirectional)	Transformer processes the full sentence at once
Vector type	Sparse	Dense	Dense	Encoding has mostly zeros, NN/Transformer give compact representations
Same word embedding	Same vector always	Same vector always	Different per context	Transformer gives dynamic embeddings and handles multiple meanings
Performance	Weak	Good	Best	Transformer uses attention and is trained on large-scale data

How to select best embedding model

1. Embedding Quality (Most Important): Check benchmark performance:

Look at:

- MTEB leaderboard
- BEIR benchmark: https://arxiv.org/abs/2104.08663

2. Dimensionality

Example:

- 384 dim → lightweight
- 768 dim → balanced
- 1536 dim → high quality but heavy

Higher dimension:

- Better semantic representation
- But more storage cost
- More memory usage
- Speed impact

3. Cost

Close source (Open AI embedding):

- Easy, high quality, API cost

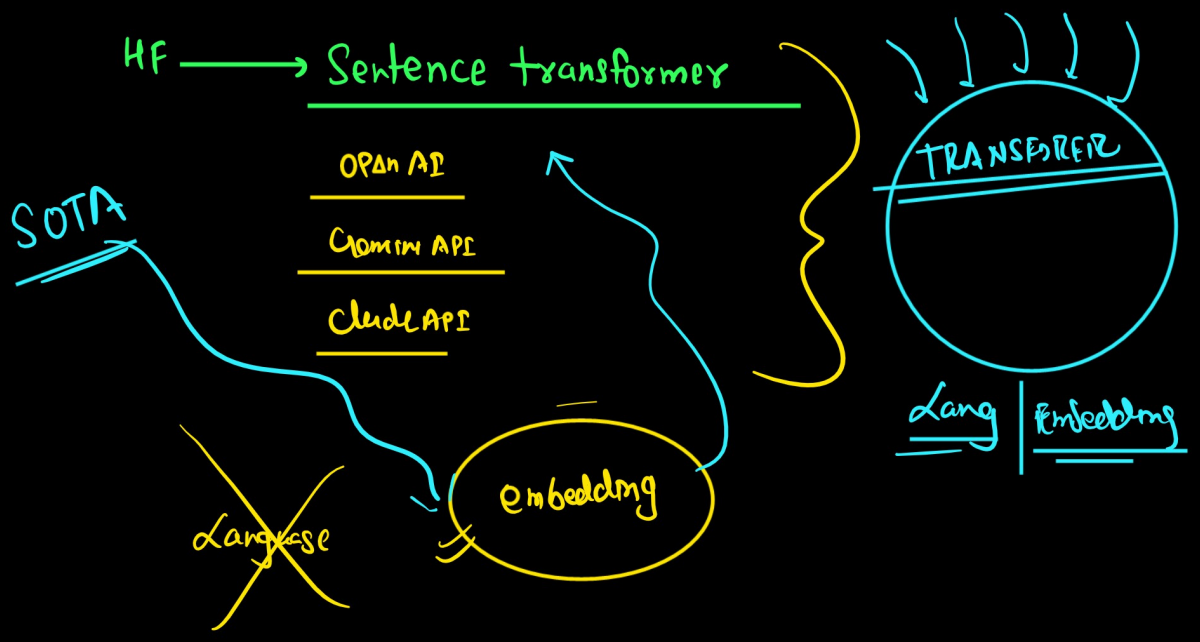
Open source:

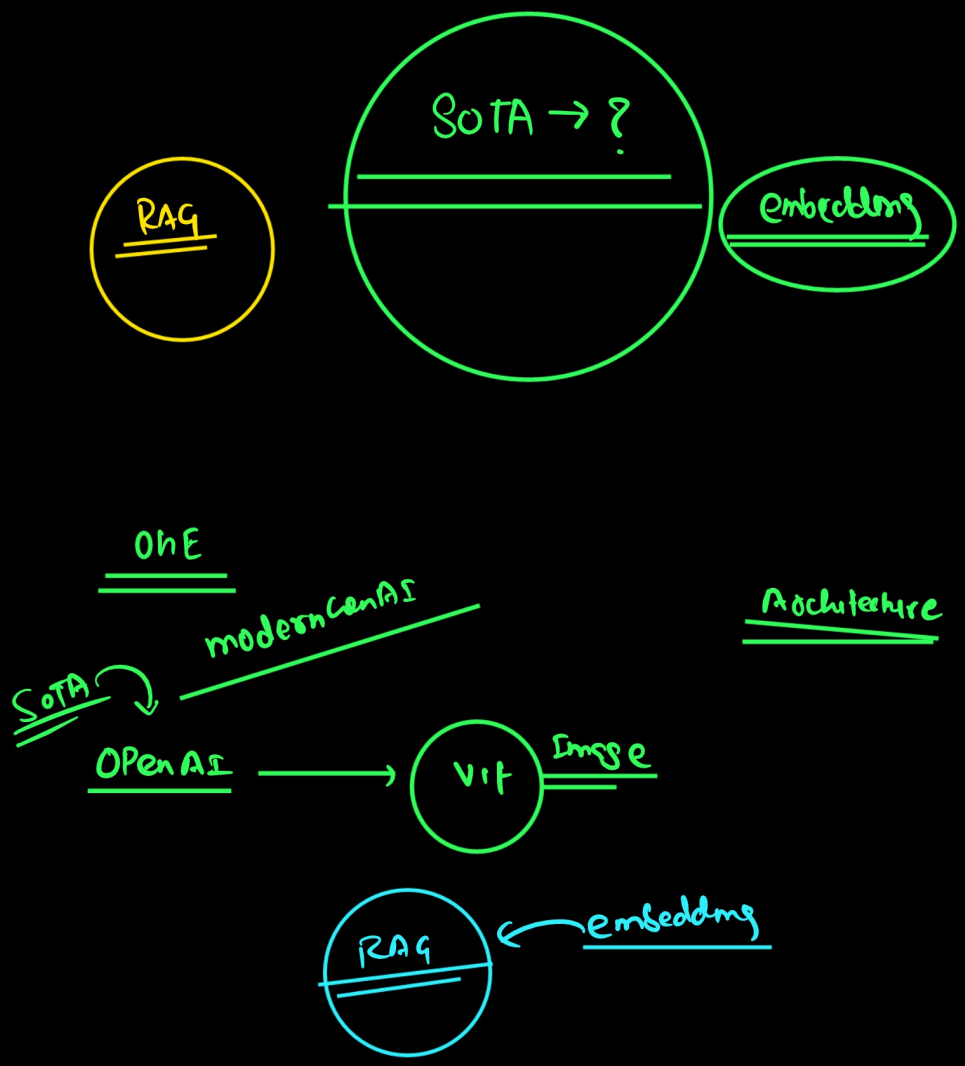
- Free but infra + scaling cost

4. Domain Suitability

General model:

- all-MiniLM
- OpenAI embedding
- Gemini embedding




SOTA → ?
RAG
embedding
OHE
modernconAI
SOTA
OpenAI
ViT
Image
Architecture
RAG
embedding